Not All Mathematical Difficulty Is the Same

Likun Xie

Some mathematical problems are hard because the solution is difficult to find. Others are hard because we do not yet understand the mathematics needed to solve them. AI's spectacular progress on the first should not cause the public to mistake it for the whole of mathematics.

People have asked how I feel about AI's rapid progress in mathematics, given concern among some mathematicians and frequent headlines about AI solving difficult problems or mathematicians moving into the AI industry.

I do not share either the anxiety or the hype, nor do I think AI changes what I value deeply in mathematics. I am writing to offer a counterpoint to a public narrative that often uses mathematics as a benchmark for AI progress while presenting mathematical research too narrowly and uniformly.

A common picture of research is: formulate a problem, try known techniques, and find the clever argument that solves it. AI is becoming very good at parts of this process: suggesting approaches, testing examples, combining methods, and increasingly producing substantial proofs.

But this is only one kind of mathematical difficulty.

Different kinds of mathematical difficulty

Sometimes we understand the mathematics around a problem reasonably well: we know the relevant objects, theories, and techniques, and the main challenge is finding the right sequence of ideas.

$$\text{Problem} \implies \text{try techniques} \implies \text{find the right idea} \implies \text{proof.}$$

This can require great ingenuity, but it also overlaps with what AI is increasingly good at: searching possibilities, combining known methods, and carrying out technical arguments.

Other problems are difficult because we do not yet understand the surrounding mathematics deeply enough. We may be missing the right objects, language, or connections altogether.

$$\text{phenomenon} \implies \text{new viewpoint} \implies \text{new structures} \implies \text{deeper understanding} \implies \text{proof.}$$

The distinction is not simply between problem solving and theory building; the two often overlap. Deep theories can emerge from attempts to solve concrete problems, while deep problems matter precisely because solving them can force us to develop new insights.

AI may be extremely useful for problems closer to the first kind. For the second, I do not see it as being anywhere close. Mathematics still contains an abundance of beautiful questions whose difficulty lies in genuine conceptual depth.

None of this is specific to one area of mathematics, but the balance may differ across fields. Problems of the first kind may be more common or more visible in some areas, which could make those fields appear more affected by AI. But even there, conceptually deep questions remain. It would therefore be misleading to suggest that some areas of mathematics are simply more likely to disappear than others. This may help explain why AI's influence appears so uneven across mathematics.

What happens when proofs become cheaper?

There is a useful historical analogy. Once computation became cheap, mathematics did not lose its purpose because machines could calculate faster than humans. The location of mathematical value shifted. Large calculations became less interesting in themselves, while understanding the structures behind them became more important.

AI may produce a similar shift at a different level. If certain kinds of technical difficulty become easier to overcome, research may move further toward conceptual depth and more fundamental advances—the direction I find most elegant and important.

More room for conceptually deep and less visible mathematics

There is also a broader consequence that I find interesting. Having interacted with mathematicians across different areas, I have become very aware of how differently various kinds of mathematics are perceived, and how some areas that I find especially elegant and deep can be comparatively underappreciated.

The current academic system naturally favors contributions that are easy to make visible: a new theorem, an improved bound, another case of a conjecture, a difficult calculation, or a clever application of an established technique. Such contributions are relatively easy to state, referee, count as papers, and explain in hiring or grant applications.

More exploratory and structural work can be harder to accommodate. Someone may spend a long time finding the right framework, understanding a collection of phenomena, connecting different theories, or developing a language that eventually makes many individual results transparent and helps solve central problems. The eventual payoff can be enormous, but the intermediate stages are often less visible as conventional research output.

If AI makes technically competent theorem production much cheaper, this balance could eventually change. There is no guarantee that academia will respond in this way. In the short term, the opposite may happen: AI could produce more papers, more results, and even higher expectations for productivity. That would be an unfortunate transitional equilibrium.

But such an equilibrium becomes harder to sustain if theorem production becomes abundant. Once large quantities of technically correct, incremental results can be generated cheaply, their sheer number becomes a poor measure of mathematical contribution. This may create more room for work whose value lies not simply in adding another theorem, but in changing how we understand a subject.

A larger landscape for individual mathematicians

AI may also change the scale at which an individual mathematician can explore ideas. Today, conceptual breadth is expensive: pursuing a connection between distant subjects may require mastering large amounts of unfamiliar machinery before the idea can even be tested.

A capable mathematical AI could lower these barriers without supplying the central idea itself. It could help navigate literature, explain unfamiliar techniques, test examples, search for obstructions, verify lemmas, and handle technical bookkeeping.

If so, an individual researcher could sustain a much broader theoretical program. AI would become not merely a faster tool, but a larger mathematical landscape.

What mathematics do we want?

If the goal were simply to maximize the number of difficult propositions proved, then sufficiently powerful AI would indeed create something close to an existential problem for mathematicians.

But I do not think that is why mathematics is meaningful.

We want to understand why phenomena occur, uncover hidden structures, find the right language, recognize connections between things that initially looked unrelated, and use difficult problems to expose the limits of our present understanding.

Proof is indispensable but proof production is not the whole purpose.

Indeed, for the deepest mathematics, proving and understanding often cannot be separated. A great problem is sometimes valuable precisely because attempting to solve it forces mathematics to acquire concepts that it did not previously have.

If AI makes routine reasoning, technical execution, and some forms of proof search dramatically cheaper, it may free more attention for problems whose difficulty is genuinely conceptual, and give individual mathematicians more room and support to pursue broad and structurally demanding ideas, whose solutions do not merely add another theorem, but change what we understand.

In that sense, AI could make pure mathematics less dominated by the search for the next clever argument and more centered on the deeper reason we solve hard problems in the first place: to discover mathematics that we do not yet know how to see.